

Probability basics

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Defining probabilities

Definition 0.1 (Sample space). The **sample space** of a random experiment, denoted Ω , is the set of all of its possible outcomes.

Example 0.1 (Sample space of a die roll). Rolling a six-sided die once has sample space $\Omega = \{1, 2, 3, 4, 5, 6\}$.

Definition 0.2 (Event). An **event** is a subset of the **sample space** Ω : a set of outcomes. An event A **occurs** when the experiment's outcome is in A .

When Ω is uncountable, such as an interval of real numbers, only the subsets in a designated collection $\mathcal{S}$ (a **σ -algebra**: a collection that contains Ω and is closed under complements and countable unions) count as events; every set that arises in these notes is in $\mathcal{S}$.

Example 0.2 (Rolling an even number). In Example 0.1, the event “the roll is even” is $A = \{2, 4, 6\}$.

Definition 0.3 (Complement of an event). The **complement** of an **event** A , denoted $\neg A$, is the event that A does not occur: the set of outcomes in the sample space Ω that are not in A .

$$\neg A \stackrel{\text{def}}{=} \Omega \setminus A$$

Other sources write A^c or $\bar{A}$.

Example 0.3 (Rolling an odd number). In Example 0.2, the complement of the event “the roll is even”, $A = \{2, 4, 6\}$, is the event “the roll is odd”, $\neg A = \{1, 3, 5\}$.

Definition 0.4 (Mutually exclusive events). Finitely or countably many events $A_1, A_2, \dots$ are **mutually exclusive** (also called **disjoint**, **pairwise disjoint**, or **mutually disjoint**) when no two of them share an outcome:

$$A_i \cap A_j = \emptyset \quad \text{for all } i \neq j$$

Example 0.4 (Low and high rolls). In Example 0.1, the events “the roll is at most 2”, $\{1, 2\}$, and “the roll is at least 5”, $\{5, 6\}$, are mutually exclusive: no roll is in both. The events “the roll is even”, $\{2, 4, 6\}$, and “the roll is at most 2”, $\{1, 2\}$, are not mutually exclusive: a roll of 2 is in both.

Definition 0.5 (Partition of an event). A **partition** of an event A is a finite or countably infinite collection of **mutually exclusive** events $A_1, A_2, \dots$ whose union is A :

$$\bigcup_i A_i = A$$

A partition of the **sample space** is the case $A = \Omega$.

Some sources also require each piece A_i to contain at least one outcome.

Example 0.5 (Partitioning the die roll). In Example 0.1, the events “the roll is even”, $\{2, 4, 6\}$, and “the roll is odd”, $\{1, 3, 5\}$, are mutually exclusive, and their union is Ω , so they partition the sample space. So do the six single-outcome events $\{1\}, \{2\}, \dots, \{6\}$. The events $\{1, 2\}$ and $\{5, 6\}$ from Example 0.4 do not partition Ω : their union leaves out 3 and 4.

Definition 0.6 (Finite additivity). A function μ that assigns a real number $\mu(A)$ to each event A is **finitely additive** if, for every finite collection of **mutually exclusive** events $A_1, \dots, A_n$, the value of their union is the sum of their values:

$$\mu(A_1 \cup \dots \cup A_n) = \sum_{i=1}^n \mu(A_i)$$

Example 0.6 (Counting outcomes is finitely additive). For the die roll in Example 0.1, let $\mu(A) \stackrel{\text{def}}{=} |A|$, the number of outcomes in A . The events $\{1, 2\}$ and $\{5, 6\}$ are mutually

exclusive (Example 0.4), and:

$$\begin{aligned}
 \mu(\{1, 2\} \cup \{5, 6\}) &= \mu(\{1, 2, 5, 6\}) && \text{(take the union)} \\
 &= 4 && \text{(count the outcomes)} \\
 &= 2 + 2 && \text{(write 4 as a sum)} \\
 &= \mu(\{1, 2\}) + \mu(\{5, 6\}) && \text{(count each event's outcomes)}
 \end{aligned}$$

The same holds for any mutually exclusive events, because the sizes of disjoint sets add, so μ is finitely additive.

Definition 0.7 (Countable additivity). A function μ that assigns a real number $\mu(A)$ to each event A is **countably additive** (also called **σ -additive**) if, for every sequence of mutually exclusive events $A_1, A_2, \dots$, the value of their union is the sum of their values:

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i)$$

Countable additivity implies finite additivity. Applying countable additivity to $\emptyset, \emptyset, \dots$, which are mutually exclusive with union $\emptyset$, gives $\mu(\emptyset) = \sum_{i=1}^{\infty} \mu(\emptyset)$, a sum of identical terms that is finite only if $\mu(\emptyset) = 0$. Then, for mutually exclusive $A_1, \dots, A_n$, set $A_{n+1} = A_{n+2} = \dots = \emptyset$ and apply countable additivity to the whole sequence. The converse fails: there exist set functions that satisfy finite additivity but fail countable additivity (see [Wikipedia: Sigma-additive set function — An additive function which is not \$\sigma\$ -additive](#)).

Example 0.7 (Counting outcomes is countably additive). For the counting function $\mu(A) \stackrel{\text{def}}{=} |A|$ of Example 0.6, take any sequence of mutually exclusive events $A_1, A_2, \dots$. No two of them share an outcome, and Ω has only six outcomes, so at most six of the A_i contain any outcomes; the rest are $\emptyset$, with $\mu(\emptyset) = 0$. The infinite sum therefore has at most six nonzero terms, and finite additivity (Example 0.6) shows that those terms add up to μ of the union. So μ is countably additive.

Definition 0.8 (Probability measure). A **probability measure** on a sample space Ω , often denoted $\Pr()$ or $P()$, is a function that assigns a number $\Pr(A)$ to each event A and satisfies:

1. For any event A , $\Pr(A) \geq 0$.
2. The probability of the whole sample space is 1:

$$\Pr(\Omega) = 1$$

3. Pr is **countably additive**: for any **mutually exclusive** events $A_1, A_2, \dots$,

$$\Pr\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \Pr(A_i)$$

Property 3 implies that Pr is also **finitely additive**. Requiring countable additivity, not just finite additivity, enables results such as the continuity of probability (if $A_1 \supseteq A_2 \supseteq \dots$ with $\bigcap_i A_i = \emptyset$, then $\Pr(A_i) \rightarrow 0$), and it is what makes sums over countably infinite **partitions** valid.

Example 0.8 (Probability measure for a fair die). For the die roll in Example 0.1, define $\Pr(A) \stackrel{\text{def}}{=} |A|/6$, where $|A|$ is the number of outcomes in A . This function is non-negative, gives $\Pr(\Omega) = 6/6 = 1$, and is **countably additive**, because it is 1/6 times the counting function of Example 0.7. The event “the roll is even” from Example 0.2 has probability $\Pr(\{2, 4, 6\}) = 3/6 = 1/2$.

Theorem 0.1 (Probability of a subset’s intersection). *If A and B are events and $A \subseteq B$, then $\Pr(A \cap B) = \Pr(A)$.*

i Proof

Proof. Since $A \subseteq B$, every outcome in A is also in B , so $A \cap B = A$, and therefore $\Pr(A \cap B) = \Pr(A)$. □

Theorem 0.2 (An event and its complement sum to 1). *For any event A and its **complement** $\neg A$:*

$$\Pr(A) + \Pr(\neg A) = 1$$

i Proof

Proof. The events A and $\neg A$ are disjoint, and their union is Ω .

$$\begin{aligned} \Pr(A) + \Pr(\neg A) &= \Pr(A \cup \neg A) && \text{(additivity of probability for disjoint events)} \\ &= \Pr(\Omega) && (A \cup \neg A = \Omega) \\ &= 1 && \text{(probability of the sample space is 1)} \end{aligned}$$

□

Corollary 0.1 (Complement rule). *For any event A :*

$$\Pr(\neg A) = 1 - \Pr(A)$$

i Proof

Proof. Subtract $\Pr(A)$ from both sides of Theorem 0.2. □

Corollary 0.2 (Complement rule in probability (π) notation). *If the probability of an event A is $\Pr(A) = \pi$, then the probability that A does not occur is:*

$$\Pr(\neg A) = 1 - \pi$$

i Proof

Proof.

$$\begin{aligned}\Pr(\neg A) &= 1 - \Pr(A) && \text{(complement rule)} \\ &= 1 - \pi && \text{(substitute } \Pr(A) = \pi \text{)}\end{aligned}$$

□

Example 0.9 (Probability of not rolling a six). For a fair die, the event “the roll is a six” has probability $\pi = 1/6$, so by Corollary 0.2 the probability of not rolling a six is $1 - 1/6 = 5/6$.

Conditional probability

Definition 0.9 (Conditional probability). For two events A and B with $\Pr(B) > 0$, the **conditional probability** of A given B , denoted $\Pr(A \mid B)$, is:

$$\Pr(A \mid B) \stackrel{\text{def}}{=} \frac{\Pr(A \cap B)}{\Pr(B)}$$

Example 0.10 (Rolling a six, given an even roll). For a fair die (Example 0.8), let $A = \{6\}$ and $B = \{2, 4, 6\}$. Then $A \cap B = \{6\}$, so:

$$\begin{aligned}
\Pr(A \mid B) &= \frac{\Pr(A \cap B)}{\Pr(B)} && \text{(definition of conditional probability)} \\
&= \frac{1/6}{3/6} && \text{(substitute the probabilities)} \\
&= \frac{1}{3} && \text{(simplify)}
\end{aligned}$$

Theorem 0.3 (Law of conditional probability). *For any two events A and B with $\Pr(B) > 0$:*

$$\Pr(A \cap B) = \Pr(A \mid B) \cdot \Pr(B)$$

i Proof

Proof. Rearranging Definition 0.9:

$$\begin{aligned}
\Pr(A \mid B) &= \frac{\Pr(A \cap B)}{\Pr(B)} && \text{(definition of conditional probability)} \\
\Pr(A \cap B) &= \Pr(A \mid B) \cdot \Pr(B) && \text{(multiply both sides by } \Pr(B))
\end{aligned}$$

□

Example 0.11 (Applying the law of conditional probability). Suppose 30% of adults exercise regularly ($\Pr(E) = 0.30$), and among adults who exercise regularly, 60% have low blood pressure ($\Pr(L \mid E) = 0.60$).

Then, by Theorem 0.3, the probability that a randomly selected adult both exercises regularly and has low blood pressure is:

$$\begin{aligned}
\Pr(L \cap E) &= \Pr(L \mid E) \cdot \Pr(E) && \text{(law of conditional probability)} \\
&= 0.60 \cdot 0.30 && \text{(substitute the given values)} \\
&= 0.18 && \text{(multiply)}
\end{aligned}$$

Theorem 0.4 (Law of total probability). *If $B_1, B_2, \dots$ is a **partition** of the sample space, with $\Pr(B_i) > 0$ for every i , then for any event A :*

$$\Pr(A) = \sum_i \Pr(A \mid B_i) \cdot \Pr(B_i)$$

i Proof

Proof. Since $B_1, B_2, \dots$ partition the sample space, the events $A \cap B_1, A \cap B_2, \dots$ are mutually exclusive and their union is A . By countable additivity (Definition 0.7), and then by Theorem 0.3:

$$\begin{aligned}\Pr(A) &= \sum_i \Pr(A \cap B_i) && \text{(countable additivity for partition of } A\text{)} \\ &= \sum_i \Pr(A \mid B_i) \cdot \Pr(B_i) && \text{(law of conditional probability; } \Pr(B_i) > 0\text{)}\end{aligned}$$

□

Theorem 0.5 (Bayes' theorem). *For any two events A and B with $\Pr(A) > 0$ and $\Pr(B) > 0$:*

$$\Pr(A \mid B) = \frac{\Pr(B \mid A) \cdot \Pr(A)}{\Pr(B)}$$

i Proof

Proof. By Definition 0.9 and Theorem 0.3:

$$\begin{aligned}\Pr(A \mid B) &= \frac{\Pr(A \cap B)}{\Pr(B)} && \text{(definition of conditional probability)} \\ &= \frac{\Pr(B \cap A)}{\Pr(B)} && \text{(intersection is commutative: } A \cap B = B \cap A\text{)} \\ &= \frac{\Pr(B \mid A) \cdot \Pr(A)}{\Pr(B)} && \text{(law of conditional probability applied to } \Pr(B \cap A)\text{)}\end{aligned}$$

□

Example 0.12 (Positive predictive value of a medical test). Suppose a disease test has 99% sensitivity and 99% specificity, and the prevalence of the disease in the population is 7%.

Let D be the event “person has the disease” and $+$ be the event “test is positive”. Then:

- $\Pr(+ \mid D) = 0.99$ (sensitivity)
- $\Pr(\neg+ \mid \neg D) = 0.99$ (specificity), so the false positive rate is $\Pr(+ \mid \neg D) = 1 - 0.99 = 0.01$
- $\Pr(D) = 0.07$ (prevalence)

By Theorem 0.5, with the denominator expanded by the [law of total probability](#) over the

partition $\{D, \neg D\}$:

$$\begin{aligned}\Pr(D \mid +) &= \frac{\Pr(+ \mid D) \cdot \Pr(D)}{\Pr(+)} && \text{(Bayes' theorem)} \\ &= \frac{\Pr(+ \mid D) \cdot \Pr(D)}{\Pr(+ \mid D) \cdot \Pr(D) + \Pr(+ \mid \neg D) \cdot \Pr(\neg D)} && \text{(law of total probability)} \\ &= \frac{0.99 \cdot 0.07}{0.99 \cdot 0.07 + 0.01 \cdot 0.93} && \text{(substitute the given values)} \\ &= \frac{0.0693}{0.0693 + 0.0093} && \text{(multiply each term in the numerator and denominator by 0.01)} \\ &= \frac{0.0693}{0.0786} && \text{(add the denominator's two terms)} \\ &\approx 0.88 && \text{(divide)}\end{aligned}$$

Even with a highly accurate test (99% sensitive and 99% specific), only about 88% of people who test positive actually have the disease, because the disease prevalence is relatively low (7%).